

Documentation - Cyclistic Capstone Project - Power BI

1. General Project Information

Dashboard Name: Cyclistic Capstone Project

Area / Department: Marketing

Company / Client: Cyclistic

Developed By: Luis Miguel

Main Stakeholders: N/A

Creation Date: February 21, 2026

Last Update: March 21, 2026

Dashboard Version: v1.0

Status: Production

General Description:

This dashboard is designed to support both the marketing and management teams by providing a comprehensive report on bicycle usage in the city of Chicago, along with key indicators related to customer behavior patterns.

2. Dashboard Objective

Primary Objectives:

- Monitor bicycle usage in the city of Chicago
- Evaluate user behavior patterns
- Support executive decision-making for product and marketing teams;

Business Questions Answered by the Dashboard:

- What are the usage patterns of bicycles based on user types?
 - Which stations have the best performance metrics and KPIs?
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4. Data Architecture Overview

Connection Type:

- Import
- DirectQuery

Refresh Frequency:

- Manual

Data Flow (Summary):

Data collection via API → ETL (Python) → Jupyter Notebook → Data Modeling → Dashboard

Tools Used:

- Power BI Desktop
 - Power BI Service
 - API
 - Python
 - Jupyter Notebook
 - n8n
-

5. Fontes de Dados

Source Name	Type	Origin	Refresh Frequency	Notes
202008-divvy-trip data	csv	API Kaggle	Daily	Future implementation of SQL database update automation

Source Details:**Data Source 1**

Name: 202008-divvy-tripdata

Type: API

Location: Kaggle

Access Credentials: Kaggle API Key

Data Format: CSV

Average Data Volume: Approximately 622,000 records

Notes: Future implementation of an automated SQL database update system

6. ETL / ELT Process

Tools Used:

- Python (Data extraction)
- Jupyter Notebook (Data processing and cleaning)
- n8n (Data collection and processing automation)

Process Description:

- Extraction: Python script responsible for downloading the data.
- Transformation: Python processes the data and stores it in the appropriate analysis directory.
- Cleaning: Jupyter Notebook performs data cleaning operations.
- Enrichment: Jupyter Notebook creates new features based on business rules.

Data Quality Validations:

- Duplicate removal;
 - Null value handling;
 - Outlier removal;
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7. Data Modeling

Model Type:

- Star Schema

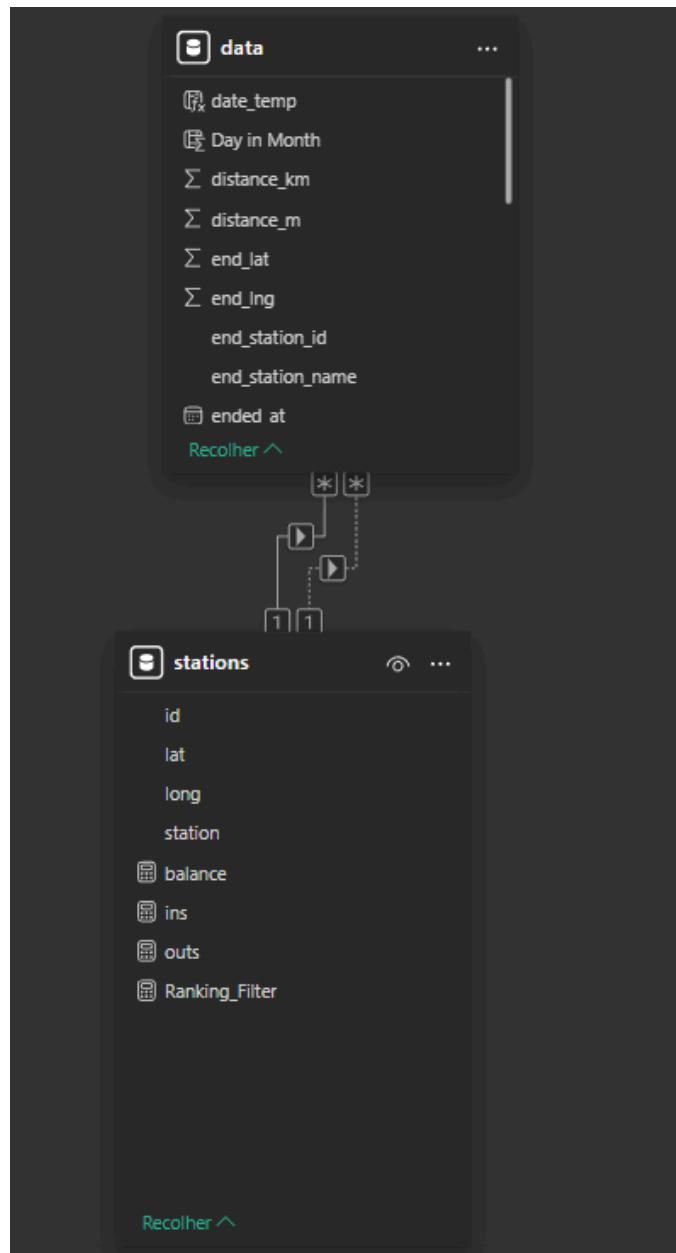
Model Description:

Two main tables are used:

data — Stores the primary trip records

stations — Stores bicycle station records

Data Model Diagram:



8. Model Tables

Table Name	Type	Description	Granularity	Source
data	Fact	Table that stores user ride records	Day, User, Station	CSV file
stations	Dimension	Table that stores the list of stations	Station level	CSV file

9. Data Dictionary

Table	Field	Data Type	Description
data	date_temp	Date	Base date used for time-based analysis
data	Day in Month	Integer	Day of the month extracted from the date
data	distance_km	Decimal	Distance traveled in kilometers
data	distance_m	Integer	Distance traveled in meters
data	end_lat	Decimal	Latitude of trip end location
data	end_lng	Decimal	Longitude of trip end location
data	end_station_id	String	End station identifier
data	end_station_name	String	End station name
data	ended_at	DateTime	Trip end date and time
data	hour_end	Integer	Trip end hour
data	hour_start	Integer	Trip start hour
data	is_weekend	Boolean	Indicates whether the trip occurred on a weekend
data	member_casual	String	User type (member or casual)
data	ride_duration	Decimal	Total trip duration (original unit)

data	ride_duration_minutes	Decimal	Trip duration in minutes
data	ride_id	String	Unique trip identifier
data	rideable_type	String	Vehicle type used (e.g., docked or electric)
data	round_trip	Boolean	Indicates whether the trip started and ended at the same station
data	start_lat	Decimal	Latitude of trip start location
data	start_lng	Decimal	Longitude of trip start location
data	start_station_id	String	Start station identifier
data	start_station_name	String	Start station name
data	started_at	DateTime	Trip start date and time
data	trip	String	Text indicating "[Start Station] => [End Station]"
data	week_day	String	Day of the week name
data	weekday_num	Integer	Day of the week number (1–7)
stations	id	Integer	Unique station identifier
stations	lat	String	Station latitude
stations	long	String	Station longitude
stations	station	String	Station name

10. Measures and KPIs (DAX)

Nome da Medida	Fórmula DAX	Descrição	Responsável pela criação
avg ride distance	AVERAGEX(data , data[distance_m])	Average distance traveled per trip (in meters)	Luís Miguel
avg ride distance (str)	FORMAT([avg ride distance] , ".00") & "m"	Average distance formatted as text for display in visuals	Luís Miguel
avg rides per day	var vMaxDate = MAX(data[started_at]) var vMinDate = MIN(data[started_at]) var vDays = DATEDIFF(vMinDate , vMaxDate , DAY)+1 RETURN DIVIDE([rides] , vDays , 0)	Average number of trips made per day	Luís Miguel
avg time ride	AVERAGEX(data , data[ride_duration_minutes])	Average trip duration	Luís Miguel

avg time ride HHMMSS	VAR TotalMinutos = [avg time ride] VAR TotalSegundos = TotalMinutos * 60 VAR Horas = INT(TotalSegundos / 3600) VAR Minutos = INT(MOD(TotalSegundo s, 3600) / 60) VAR Segundos = INT(MOD(TotalSegundo s, 60)) RETURN FORMAT(Horas, "00") & ":" & FORMAT(Minutos, "00") & ":" & FORMAT(Segundos, "00")	Average trip time formatted in HH:MM:SS format	Luís Miguel
Destaque Fim de Semana	IF(SELECTEDVALUE('dat a'[is_weekend]) = "True", [rides] * 1.5,	Indicator highlighting weekend performance (NOT IN USE)	Luís Miguel

	0)		
distance total	SUMX(data , data[distance_m])	Total sum of distances traveled	Luís Miguel
ins_	COUNTROWS(FILTER(data, [start_station_id] = [end_station_id]))	Number of rides that start and end at the same station (NOT IN USE)	Luís Miguel
rate of return	COUNTROWS(FILTER(data, [start_station_id] = [end_station_id])) / [rides]	User return rate to a station	Luís Miguel
rides	DISTINCTCOUNT(data[ride_id])	Total number of trips	Luís Miguel
rides_same_place_ye s_or_no	var vIncludeSamePlace = [rides] var vNoIncludeSamePlace = CALCULATE([rides] , data[round_trip]="differe nt_place") RETURN IF(SELECTEDVALUE(filter _Place[Same_place_co d]) = 0 ,	Indicator of whether the trip started and ended at the same location	Luís Miguel

	vNoIncludeSamePlace , vIncludeSamePlace)		
all rides day in month	CALCULATE([rides] , ALL(data[Day in Month]))	Total trips with date filters removed	Luís Miguel
all rides hour_start	CALCULATE([rides] , ALL(data[hour_start]))	Total trips with date/time filters removed	Luís Miguel
pareto day in month	DIVIDE([rides cumulative Day in Month] , [all rides day in month] , 0)	Cumulative percentage of trips per day of the month (Pareto analysis)	Luís Miguel
pareto hour_start	DIVIDE([rides cumulative hour_start] , [all rides hour_start] , 0)	Cumulative percentage of trips per start time (Pareto analysis)	Luís Miguel
rides cumulative Day in Month	CALCULATE([rides], FILTER(	Total cumulative trips throughout the days of the month	Luís Miguel

	ALLSELECTED('data'[Day in Month]), ISONORAFTER('data'[Day in Month], MAX('data'[Day in Month]), DESC)))		
rides cumulative hour_start	CALCULATE([rides], FILTER(ALLSELECTED('data'[Day in Month]), ISONORAFTER('data'[Day in Month], MAX('data'[Day in Month]), DESC)))	Total cumulative trips throughout the hours of the day	Luís Miguel
balance	[ins] - [outs]	Balance of rides per station	Luís Miguel
ins	CALCULATE(COUNTROWS(data),	Number of rides that arrive at a specific station	Luís Miguel

	<pre> USERRELATIONSHIP(st ations[station] , data[end_station_name])) </pre>		
outs	COUNTROWS(data)	Number of rides that depart at a specific station	Luís Miguel
Ranking_Filter	<pre> VAR RankGanham = RANKX(ALL('stations'), [balance], , DESC) VAR RankPerdem = RANKX(ALL('stations'), [balance], , ASC) RETURN IF(RankGanham <= 5 RankPerdem <= 5, 1, 0) </pre>	A measure that serves as a filter for the "Bike Balance" graph, selecting only the top 5 with the most positive balance and the top 5 with the most negative balance.	Luís Miguel

Main KPIs:

- rides
 - avg ride distance
 - distance total
 - avg time ride HHMMSS
 - avg rides per day
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11. Dashboard Structure

Page list:

Page	Description	Objective	Audience
Summary	Main page with key indicators	Provide an overview of rides during the selected period	All users
User Type	Behavioral analysis of different user types	Analyze individual user metrics	All users
Bike	Performance analysis of bicycle types	Understand how bicycles are being used	All users
Stations	General station analysis	Identify most used stations and their return rates and balance	All users
Day/Date	Analysis of weekdays and time patterns	Understand user behavior patterns across days and times	All users
Pareto 80/20	Pareto analysis	Identify the time periods responsible for the majority of usage	All users

12. Security and Access (RLS)

Row-Level Security Implemented:

- No

Security Rule Description:

All users have full access to the entire dashboard.

13. Dashboard Performance

Average Refresh Time: N/A

Average Load Time: N/A

Main Optimizations Applied

- N/A
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14. Monitoring and Maintenance

Maintenance Owner: Luís Miguel

Review Frequency:

- Monthly

Maintenance Checklist:

- Check refresh failures
 - Validate data quality
 - Review performance
 - Update business rules
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15. Version Control

Version	Date	Responsible	Description
v1.0	March 21, 2026	Luís Miguel	First production deployment

16. Roadmap and Future Improvements

Planned Improvements:

- Design and implement a data architecture responsible for automating data collection and processing

Technical Backlog:

- NA
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17. Attachments

- Power BI Dashboard Link:
<https://app.powerbi.com/view?r=eyJrIjojOGUzNGU1MTUtMThmMy00MDJmLWE2MzctNWVjY2Y1MDQ3ZDEyIiwidCI6IjFiZGZlMjc4LTVmZjgtNGNkYS04MWE5LWlxMzFmM2QwZDRjMiJ9>
- Git Repository: <https://github.com/LuisMig-code/Cyclistic-Capstone-Project-Analytics>
- Architecture Diagram: NA
- API Documentation: NA